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A Novel Catalytic Hydrothermal System as a Green Approach for Sustainable Artificial Maturation of the Kerogen-Rich Low-Permeable Shale Rocks

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Abstract

The effectiveness of an iron nitrate catalyst ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$) was experimentally investigated to improve the efficiency of steam injection for *in-situ* hydrothermal conversion of Domanik low-permeable shale rock, with the aim of improving both the quantity and quality of produced synthetic oil. The experiments were performed under hydrothermal condition at 300°C and 350°C for 24 hours and at 400°C for 3 hours. The generated synthetic oils and produced gases were analyzed using GC, GC-MS chromatography, gas chromatography, elemental and SARA analysis. The obtained results revealed that the addition of iron nitrate led to a significant increase in synthetic oil yield by 2.56% at 300°C, 5.39% at 350°C and 4.87% at 400°C, as opposed to non-catalytic conditions. The optimal condition was identified at 350°C, which yielded higher amounts of light n-alkanes ($\text{C}_9\text{-C}_{20}$) compared to heavy n-alkanes. The results of SARA analysis and GC-MS also showed high aliphatic hydrocarbons content within the generated synthetic oil. Furthermore, XRD analysis revealed that the presence of iron sulfides (FeS_2) and iron oxides (Fe_2O_3) plays a significant role in facilitating the effective desulfurization of bitumen and kerogen. These findings highlight the catalytic potential of water-soluble iron nitrate in hydrothermal treatment, offering valuable insights for optimizing synthetic oil production from oil shale.

Introduction

Oil shale is a significant and largely untapped source of hydrocarbons. Like tar sands and coal, it is classified as an unconventional resource because hydrocarbons cannot be extracted directly through conventional drilling and pumping. Instead, the oil must be produced thermally using thermal methods, including *in-situ* hydrothermal treatment and pyrolysis (retorting). Oil shale contains kerogen, a solid organic material tightly embedded in the mineral matrix. Globally, oil shale deposits are abundant, offering substantial potential for hydrocarbons production (shale oil), which could contribute to the decline of conventional petroleum reserves (Speight 2012, 2016b, 2016a; Saeed, Taura, et al. 2022). These deposits vary considerably in scale and economic viability, ranging from minor, uneconomical occurrences to vast

formations spanning thousands of square kilometers and holding billions of barrels of recoverable shale oil. Currently, conventional crude oil production remains more cost-effective than shale oil due to the additional costs associated with mining and processing oil shale. Consequently, commercial exploitation of oil shale is presently limited to select operations in the USA, Estonia, Brazil and China. However, with declining global crude oil reserves and the rising cost of petroleum products, oil shale may emerge as a viable alternative to meet future fossil fuel demand (Culbertson and Pitman 1973; Bartis et al. 2005).

In recent times, the extraction of shale oil from diverse oil shale deposits has been carried out by several researchers and companies using *in-situ* conversion technology under catalytic and non-catalytic hydrothermal conditions. (Lewan, Winters, and McDonald 1979; Saeed et al. 2024; Deng et al. 2011; Saeed et al. 2025). For instance, Saeed, Shadi A., et al. investigated the effects of temperature, reaction time and the changes of structure and pore size on different Formation such as Domanik and Natih B Formations. They found that, 350°C was the optimal temperature for generating high-quality synthetic oils. Additionally, hydrothermal treatment enhanced pore size and structure, leading to a notable increase in the permeability (connectivity) and porosity of the studied Formations (Saeed, Taura, et al. 2022; Saeed et al. 2024). Similarly, Hu et al. demonstrated successful synthetic oil extraction at 350°C, with notable gas production observed at temperatures exceeding 350°C (Hu et al. 1999). In Sweden, synthetic oils from oil shale was extracted commercially by *in-situ* thermal technology in the presence of water, demonstrating industrial applicability of this approach (Assessment 1980). The application of catalyst combined with hydrothermal system further enhanced the extraction efficiency and product quality. Saeed, Shadi A., et al. (2025) demonstrated that the addition of a water-soluble Ni-based catalyst notable improved not only the quality and yield of generated synthetic oils but also enhances rock wettability (Saeed et al. 2025). Wang, Li, et al. demonstrated that nickel chloride hexahydrate significantly reduced the activation energy required for oil shale pyrolysis while enhancing overall conversion rates (Wang et al. 2024). Lu Y. et al. (2020) conducted systematic research on the catalytic pyrolysis of Changji shale rocks using various compounds. Their work revealed that specific catalysts, including sodium acetate, calcium acetate and magnesium oxide effectively enhanced oil yield and quality (Lu et al. 2020). However, determining the optimal conditions and appropriate catalysts is essential to improve the *in-situ* extraction of oil shale. Therefore, this work proposes the use of a water-soluble iron-based catalyst ($\text{FeNO}_3 \cdot 9\text{H}_2\text{O}$) to enhance the steam injection, improving the hydrothermal conversion efficiency and oil extraction from Domanik oil shale.

Experimental section

Materials and main characteristics of studied sample

The investigated samples originated from Domanik shale Formation within the Kuzminovsky region, situated in Russia's Volga-Ural Basin. This geological formation consists mainly of siliceous, carbonate-rich, and clay-bearing shales, dating back to the Late Devonian-Early Carboniferous period (Saeed, Hakimi, et al. 2022). The Domanik deposits extend across the Volga-Ural region, exhibiting significant variations in subsurface depth (Liang et al. 2020; Hakimi et al. 2023). Mineralogical characterization of the raw sample was performed using a Bruker D2 Phaser X-ray diffractometer (Germany). Quantitative analysis indicated the following composition: 53% carbonate minerals (calcite and dolomite), 34% quartz, and 12% clay minerals, with trace amounts of pyrite (Figure 1). Prior to organic geochemical analysis, the sample underwent pretreatment with 10% hydrochloric acid to remove carbonate components. Subsequent measurement of total organic carbon content using a 2400 Series II elemental analyzer yielded a value of 7.8%. Through comprehensive characterization in previous studies, the kerogen present in these samples has been identified as Type I/II marine organic matter (Saeed, Hakimi, et al. 2022).

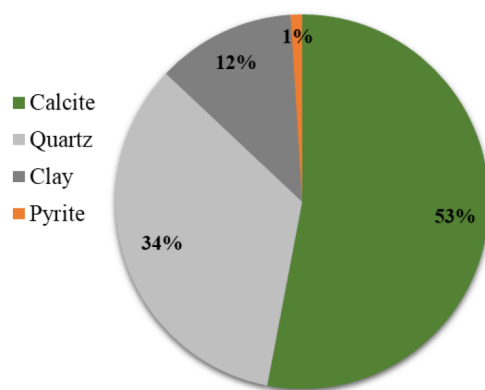


Figure 1—Mineral composition of initial Domanik rock sample by XRD

Catalyst preparation for hydrothermal treatment of Domanik rock sample

This study investigated the catalytic performance of iron nitrate $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ for the hydrothermal conversion of Domanik shale sample. The selected catalyst exhibited characteristic violet crystalline, with a measured density of 1.68 g/cm^3 when used in the hydrothermal treatment of the studied shale sample. About 13.82% of its total weight of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ is composed of iron.

Experimental procedure for the generation of synthetic oils from oil shale

Catalytic and non-catalytic hydrothermal processing experiments were conducted in a 500 mL batch reactor designed for high-pressure and high-temperature operations (Figure 2). For each trial, approximately 50 grams of pulverized oil shale sample was combined with 50 grams of deionized water, along with 1 gram of iron nitrate catalyst for the catalytic runs. This mixture was transferred to the autoclave reactor, which was then hermetically sealed. Prior to heating, the system was flushed with nitrogen gas (N_2) to create an inert atmosphere, with an initial pressure maintained at 2.1 bar. Temperature was subsequently increased to the desired range of 300, 350 and 400°C at a controlled rate of 8°C per minute (For full procedural details, please refer to our prior work (Saeed, Taura, et al. 2022)).

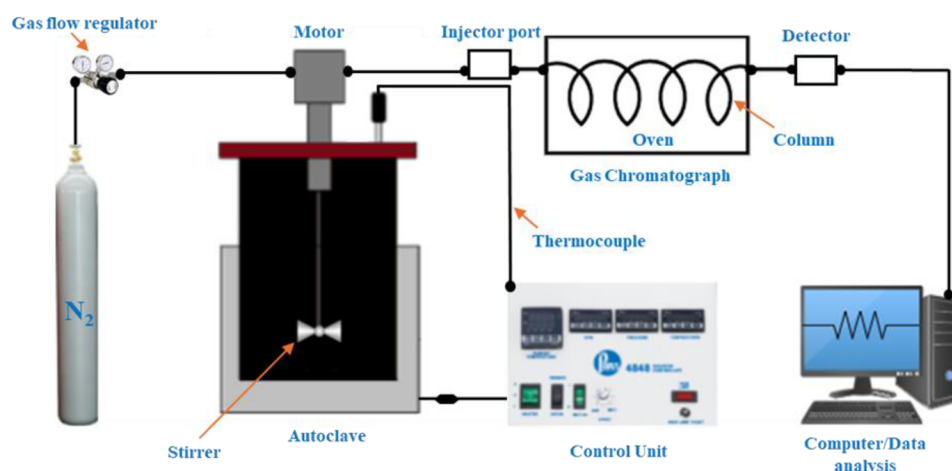
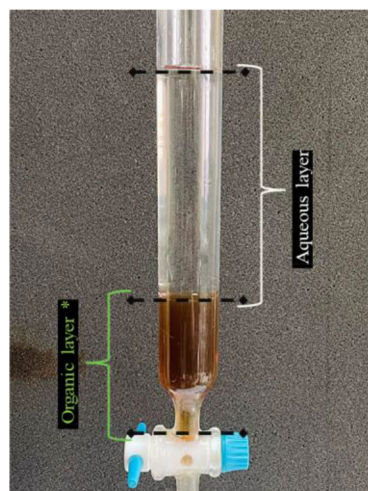


Figure 2—Schematic diagram of batch reactor integrated with gas analysis system.

Material balance during hydrothermal treatment of studied shale sample

Following the release of gaseous products, the remaining materials in the batch reactor, including spent shale, water, and synthetic oil were carefully decanted and filtered using filter paper into a pre-weighed flask. This process effectively separated the liquid phase (synthetic oil and aqueous components) from spent shale residue. Residual synthetic oil adhering to the reactor walls was subsequently recovered using a ternary

solvent mixture of isopropyl alcohol (IPA), chloroform (TCM), and toluene (TOL). For full procedural details, please refer to our prior work (Saeed, Taura, et al. 2022). Figure 3 provides a schematic of the separation methodology.



Separatory funnel

*Obtained synthetic oil-I + solvent

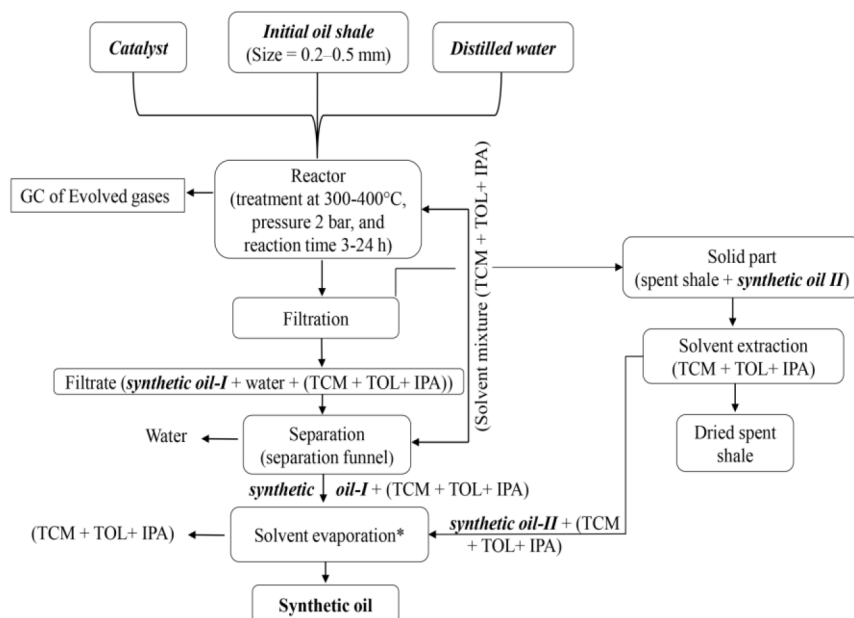


Figure 3—A scheme of the products

Analysis of the obtained products after hydrothermal treatment

GC analysis of released gases

The gaseous products released during hydrothermal conversion were examined using gas chromatography (GC). A Chromatek-Kristall 5000.2 system connected to thermal conductivity detectors and flame ionization detection (FID) was employed for this purpose. A sample of gas has been determined directly from the autoclave through a specialized transfer line equipped to the chromatograph. For detection, H_2S and CO_2 were separated using a Hayesep Q packed column (1.5 m length, 2 mm diameter, 80/100 mesh) and measured with a DTP-1 detector. Meanwhile, hydrogen, oxygen, nitrogen, and methane were resolved on an $\text{HsR} + \text{NaX}$ column (2 m length, 3 mm diameter, 60/80 mesh) and detected via TCD-2. Hydrocarbon gases were analyzed using a CR-1 PONA capillary column (100 m length, 0.25 mm diameter, 0.5 μm particle size) with FID detection. Further methodological details can be found in prior work (Al-Muntaser et al. 2025)

Gas chromatography (GC) and chromatography/mass spectrometer (GC-MS) analysis

The n-alkane distribution of synthetic oils produced during hydrothermal treatment was examined using an Agilent 7890A gas chromatograph. Additionally, a Chromatec-Crystal 5000 GC-MS system equipped with an ISQ mass selective detector (USA) was used to determine the oil composition. Data analysis was performed using Xcalibur software. The GC separation was conducted using a 30 m $\times$ 0.25 mm capillary column, employing helium as the carrier gas at a rate flow of 1 mL/min. The injector temperature was maintained at 310°C. Further experimental details can be found in earlier work (Saeed, Taura, et al. 2022; Al-Muntaser et al. 2024).

Elemental analysis

The quantitative elemental analysis of synthetic oils was performed using a Perkin Elmer 2400 Series II CHNS/O analyzer following ASTM D5291 standards. Precisely weighed samples (1.5-2.0 mg) were encapsulated in high-purity tin capsules for carbon/hydrogen analysis and aluminum capsules for nitrogen/

sulfur determination. The automated 60-position autosampler introduced samples into the combustion chamber maintained at 1100°C under continuous oxygen flow (99.999% purity). Combustion products were separated by adsorption chromatography and quantified using thermal conductivity detection, with acetanilide as calibration standard. System validation included daily blank runs and triplicate sample analyses to ensure <0.3% relative standard deviation.

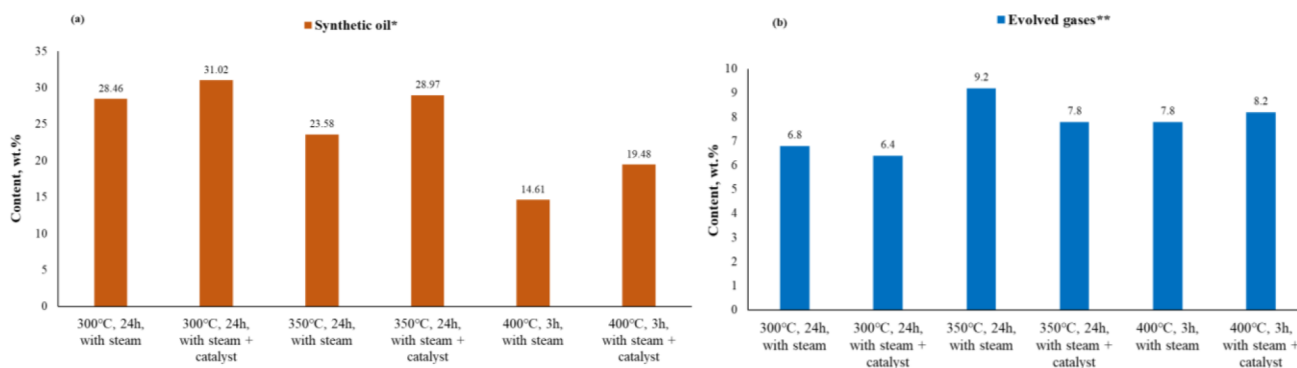
Low field nuclear magnetic resonance (LF- NMR)

Complementary molecular characterization was achieved through low-field nuclear magnetic resonance (LF-NMR) using a 20 MHz SKB Chromatec proton spectrometer (Russia). Samples were analyzed in sealed 10 mm ampoules, with measurements capturing total T2 relaxation profiles at a proton resonance frequency of 20 MHz. The instrument's receiver demonstrated a rapid response time (<10 μ s dead time), as detailed in prior work (Vladimir Y Volkov et al. 2021; Khelil et al. 2024).

Results and discussion

Material balance during hydrothermal treatment of oil shale

The distribution of obtained products after catalytic and non-catalytic experiments, including generated synthetic oils and produced gases is presented in Figure 4. Iron (III) nitrate notably enhanced the yields of synthetic oils across all tested temperatures (300-400°C), while the results of gas chromatography revealed a significant reduction in hydrocarbons gases (Table 1). The increase in the synthetic oil production is attributed to promoted ring-opening, hydrogenation, and dealkylation reactions, which simultaneously inhibit excessive cracking while converting kerogen into stable liquid hydrocarbons (Che et al. 2024; Xiaowei et al. 2021). Although the yields of methane (C_1) and light hydrocarbons (C_1 – C_4) decreased, the production of total gas increased slightly due to higher yields of non-hydrocarbons gases such as H_2S and CO_2 . The highest production of synthetic oil was recorded in catalytic experiments, especially at 350°C, which proved to be the best condition in terms of yield and high-quality oil, as evidenced by different analytical results (LF-NMR, GC-MS and GC).



* The yield of synthetic oils produced (wt. %) was calculated relative to the organic matter content (3.9%) in the initial 50 g of oil shale.

** Gas yield (wt. %) was calculated based on the mass of initial oil shale rock (50g).

Figure 4—Material balance after catalytic and non-catalytic hydrothermal conversion: (a) yield of synthetic oil and (b) produced gases yield* The yield of synthetic oils produced (wt. %) was calculated relative to the organic matter content (3.9%) in the initial 50 g of oil shale.** Gas yield (wt. %) was calculated based on the mass of initial oil shale rock (50g).

GC analysis of evolved gases

Table 1 shows the gaseous products obtained during catalytic and non-catalytic hydrothermal treatment. The results reveal that carbon dioxide (CO_2) is the primary gaseous component. Typically, CO_2 production results from the thermal decomposition of calcium carbonate (CaCO_3) (Saeed et al. 2024; V.Y. Volkov et al. 2021). However, with the addition of a catalyst, the number of CO_2 raise across all tested experiments, shifting from 44 mol.%, 40 mol.%, and 22 mol.% to 46.46 mol.%, 41.31 mol.%, and 33.12 mol.%, respectively. The slight rise in CO_2 may due to the catalyst ability to promote the thermal decomposition of kerogen, leading to the free radicals' formation (unstable molecule) and CO. The CO subsequently undergoes a water-gas shift reaction, producing more CO_2 . The increase of H_2S in the presence of iron nitrate catalyst suggests the cleavage of C-S bonds in complex organic structures such as asphaltenes and resins (Alemán-Vázquez et al. 2016; Clark, Hyne, and Tyrer 1983; Nassar, Hassan, and Pereira-Almao 2011; Zhou et al. 2024). Despite the increase in H_2S and CO_2 , the use of a catalyst led to a notable decrease in C_1 , $\sum C_2-C_4$, H_2 and CO. This decrease can be explained by hydrogenation efficiency of the catalyst, which improves the saturation of unstable molecule, and converts them into saturated hydrocarbons (Liu et al. 2009).

Table 1—Released gases after catalytic and non-catalytic hydrothermal treatment.

Compounds	Gas yields, mol.%					
	300 °C, 24h, with steam	300 °C, 24h, with steam + catalyst	350 °C, 24h, with steam	350 °C, 24h with steam + catalyst	400 °C, 3h, with steam	400 °C, 3h, with steam + catalyst
C_1	0.2044	0.199	4.201	2.872	11.05	8.621
$\sum C_2-C_4$	0.18	0.201	2.87	1.895	5.20	4.96
H_2	6.5649	1.738	10.911	8.866	11.72	10.12
H_2S	0.0091	0.005	0.0115	0.019	0.063	0.083
CO_2	44.774	46.46	40.276	41.31	22.64	33.12
CO	-	-	0.3683	0.313	0.63	0.432

Elemental analysis of initial bitumen and synthetic oils

Table 2 presents the elemental composition of initial kerogen and synthetic oil obtained from both after catalytic and non-catalytic hydrothermal treatment. The use of $\text{Fe}(\text{NO}_3)_3$ led to a rise in the H/C ratio, suggesting that the produced synthetic oils contained a high aliphatic hydrocarbons content. This observation aligns with the results of n-alkane distribution shown in Figure 5. The higher H/C ratio likely results from the ability of catalyst to facilitate the transfer of hydrogen from water, thereby stabilizing free radicals formed during thermal cracking (Al-Muntaser et al. 2021; Suwaid et al. 2020). At 350 and 400°C, the catalyst demonstrated notable removal of sulfur capability. The reduced sulfur content suggests hydrothermal desulfurization of sulfur-rich compounds within kerogen and pre-existing bitumen. According to gas composition and XRD data, most of the sulfur was expelled as H_2S and subsequently trapped by the catalyst, forming iron sulfides and iron oxides (FeS_2 and Fe_2O_3). Additionally, the produced synthetic oils exhibited higher nitrogen content, especially under catalytic conditions. This increase is likely due to the thermal stability of nitrogen-rich heterocyclic compounds, which are abundant in bitumen-kerogen structures (Bennett and Love 2000).

Table 2—Elemental analysis of initial bitumen and generated synthetic oils.

Temp. (°C)	Reaction time (h)	System	Elemental analysis, wt. %				
			C	H	N	S	N _H /N _C
Kerogen	-	-	58.77	4.86	1.21	5.35	58.77
300	24	with steam	82.55	10.56	0.24	3.4	1.53
		with steam + catalyst	80.39	10.50	0.32	3.51	1.56
350	24	with steam	82.87	10.77	0.16	3.2	1.55
		with steam + catalyst	81.04	10.96	0.54	3.16	1.62
400	3	with steam	83.06	9.36	0.08	2.97	1.61
		with steam + catalyst	80.09	10.85	0.37	2.95	162

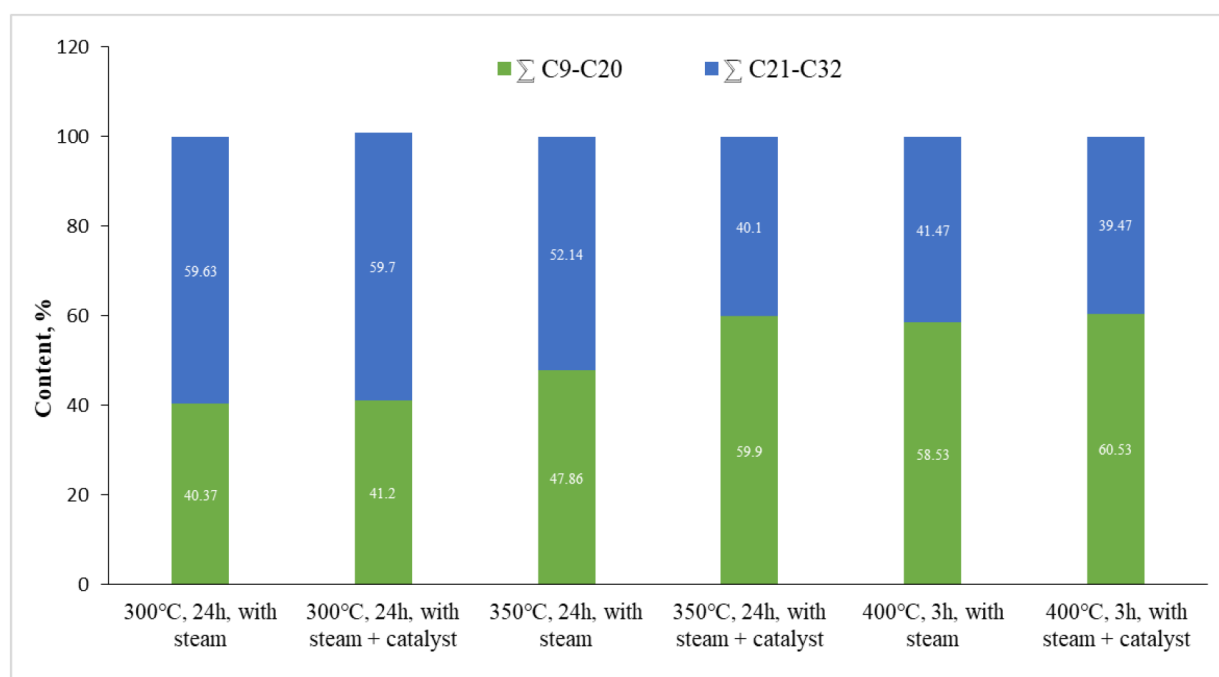


Figure 5—Carbon number distribution in synthetic oil

Carbon number distributions of synthetic oils

Figure 5 presents the n-alkanes distribution in synthetic oils produced after catalytic and non-catalytic hydrothermal treatment. The catalytic hydrothermal conversion, especially at 350°C leads to a significant increase in the number of light fractions (C₉-C₂₀) and a reduction in heavy fractions (C₂₀-C₃₂). The increase in C₉-C₂₀ can be attributed to the dealkylation of long-alkyl side chains substituents in heavy components presents in kerogen and initial bitumen within Domanik oil shale (Suwaid et al. 2023; Saeed, Taura, et al. 2022). These findings align with GC-MS and SARA analyses, demonstrating that optimal condition enhancement of the synthetic oil occurs at 350 °C. The observed changes in n-alkane distribution directly correlate with enhanced synthetic oil yields (Figure. 4a), demonstrating the catalyst's dual capability to simultaneously improve both the production of synthetic oil and hydrocarbon composition (Yi, Babadagli, and Li 2018; Al-Muntaser et al. 2020). This synergistic effect confirms that catalytic treatment not only increases extraction efficiency but also optimizes the molecular weight distribution of generated hydrocarbons.

GC–MS analysis of obtained synthetic oils

In this work, GC-MS analysis was used to study the composition of the obtained synthetic oils after catalytic and non-catalytic hydrothermal treatment of Domanik oil shale. According to the obtained results, catalytic experiments further enhanced the production of aliphatic hydrocarbons to 19.81% at 300°C, 18.16% at 350°C and 28.35% at 400°C. This increase in aliphatic hydrocarbons is accompanied by decrease in aromatic hydrocarbons, as shown in Figure 6. The observed reduction in aromatics suggests that the catalyst preferentially promotes hydrogenation and saturation rather than cyclization and aromatization (Foss et al. 2017). These results indicate that the selected catalyst system actively enhances the reaction network to favor aliphatic hydrocarbons production while inhabiting aromatic byproducts. The catalytic behavior has important implications for the generation of higher-quality synthetic oil characterized by a high H/C ratio and reduced polycyclic aromatic content (Hyne 1986).

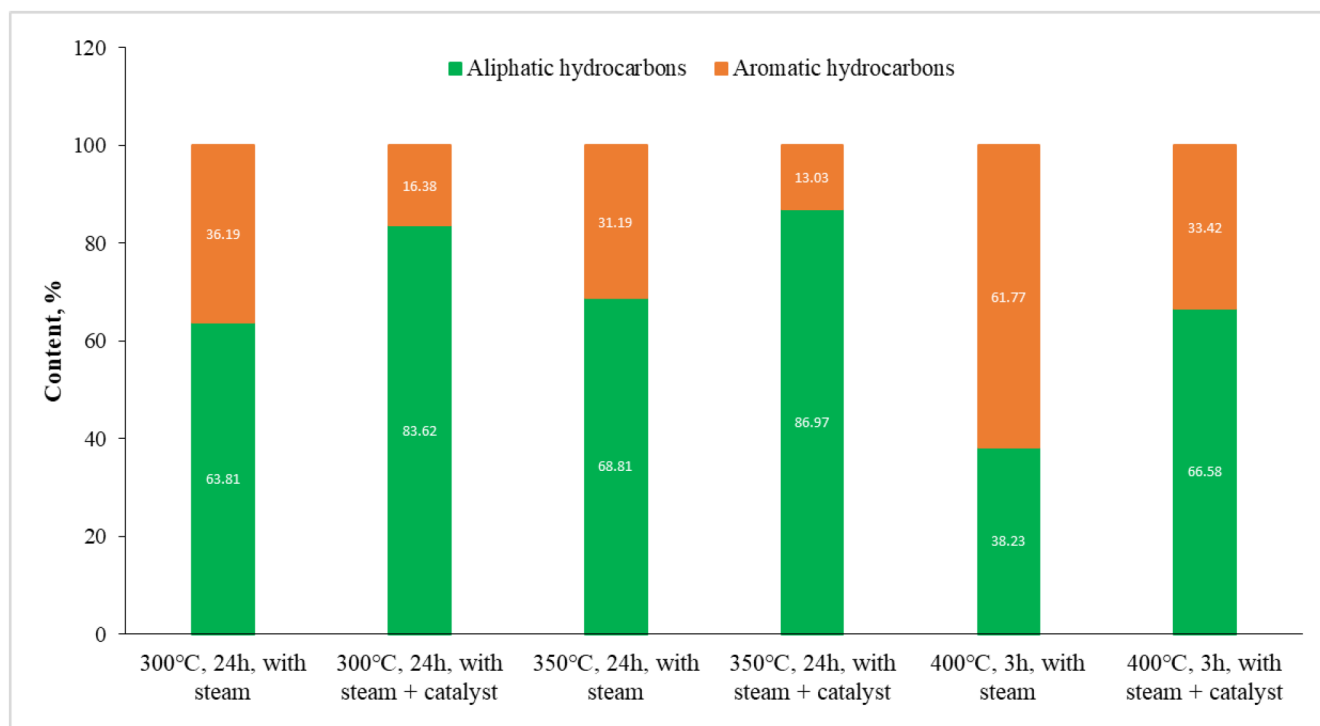


Figure 6—Synthetic oil compositions using GC–MS

LF-NMR relaxometry before and after hydrothermal treatment

The SARA analysis of obtained generated synthetic oil reveals distinct compositional shifts between catalytic and non-catalytic experiments (Figure 7). The results shows that the presence of $\text{Fe}(\text{NO}_3)_3$ significantly enhances light hydrocarbon production (saturates and aromatics) as the temperature increases, with yields rising from 28%, 46.9% and 58.4 to 35.6%, 56.2% and 69.6% respectively. This improvement in light hydrocarbons is accompanied by significantly decrease in heavy molecules (resins and asphaltenes), suggesting that the catalyst promotes the thermal cracking of resins and asphaltenes within initial bitumen and kerogen into lighter, more valuable fractions (Lai et al. 2017; Huang et al. 2018; Kang et al. 2022). These findings highlight the effectiveness of water-soluble catalyst in enhancing the hydrothermal conversion efficiency, favoring higher yields of lighter hydrocarbon and reducing undesirable heavy residues, which is critical for optimizing oil shale conversion processes.

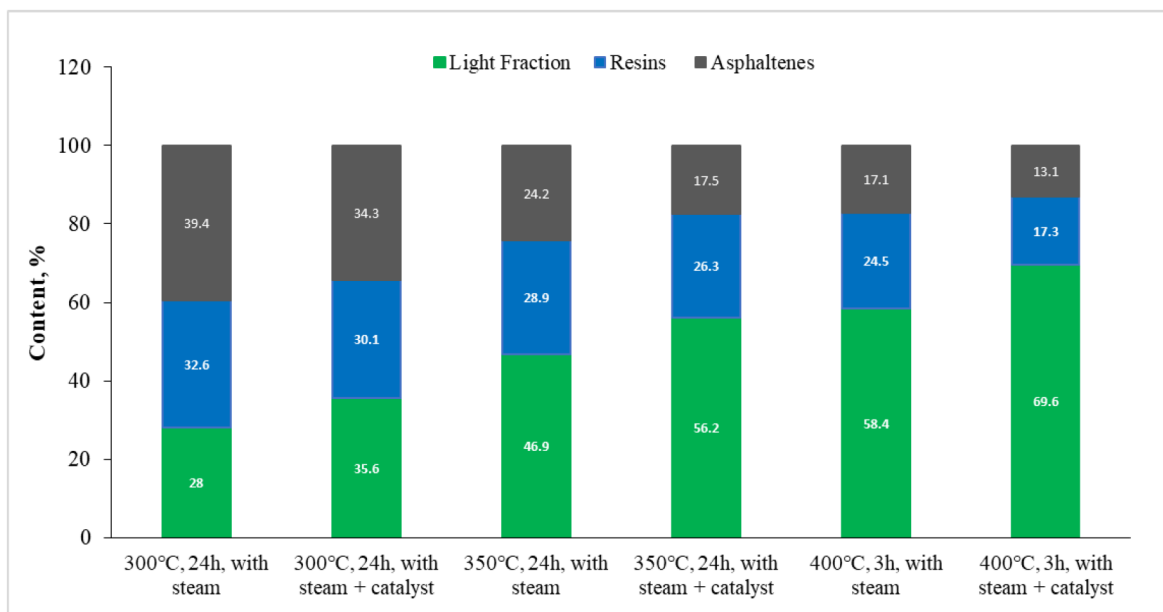


Figure 7—SARA analysis of generated synthetic oil by NMR

XRD investigation of Spent Shale during Catalytic Hydrothermal conversion

The XRD patterns of the spent shale after catalytic hydrothermal conversion in the presence of iron nitrate catalyst is shown in Figure 8. The analysis reveals the presence of crystalline phases, including CaCO_3 , MgCO_3 (ICDD No. 00-060-0473), SiO_2 (ICDD No. 01-083-2467), FeS_2 and Fe_2O_3 (ICDD No. 00-054-0489). These phases reflect the inherited mineralogy of the raw Domanik shale and the transformations induced by the catalytic process under hydrothermal conditions. The detection of iron sulfides (FeS_2) and iron oxides (Fe_2O_3) confirms that the iron nitrate catalyst undergoes transformation during the reaction, forming active metal oxide and sulfide phases. These iron-containing phases are formed as a result of the catalyst's role in sulfur removal from bitumen and kerogen macromolecules present in the Domanik oil shale. During hydrodesulfurization, organosulfur compounds are cleaved, releasing hydrogen sulfide (H_2S), which reacts with iron species to form iron sulfides and oxides nanoparticles. These nanoparticles act as active catalytic sites facilitating desulfurization, thereby enhancing the quality of the obtained synthetic oil. This transformation and catalytic behavior align with findings in literature where iron oxides and sulfides serve as active phases in hydrodesulfurization processes (Mallick, Witcomb, and Scurrall 2007). This process highlights the catalyst's efficiency in enhancing desulfurization, thereby significantly enhancing synthetic oil quality.

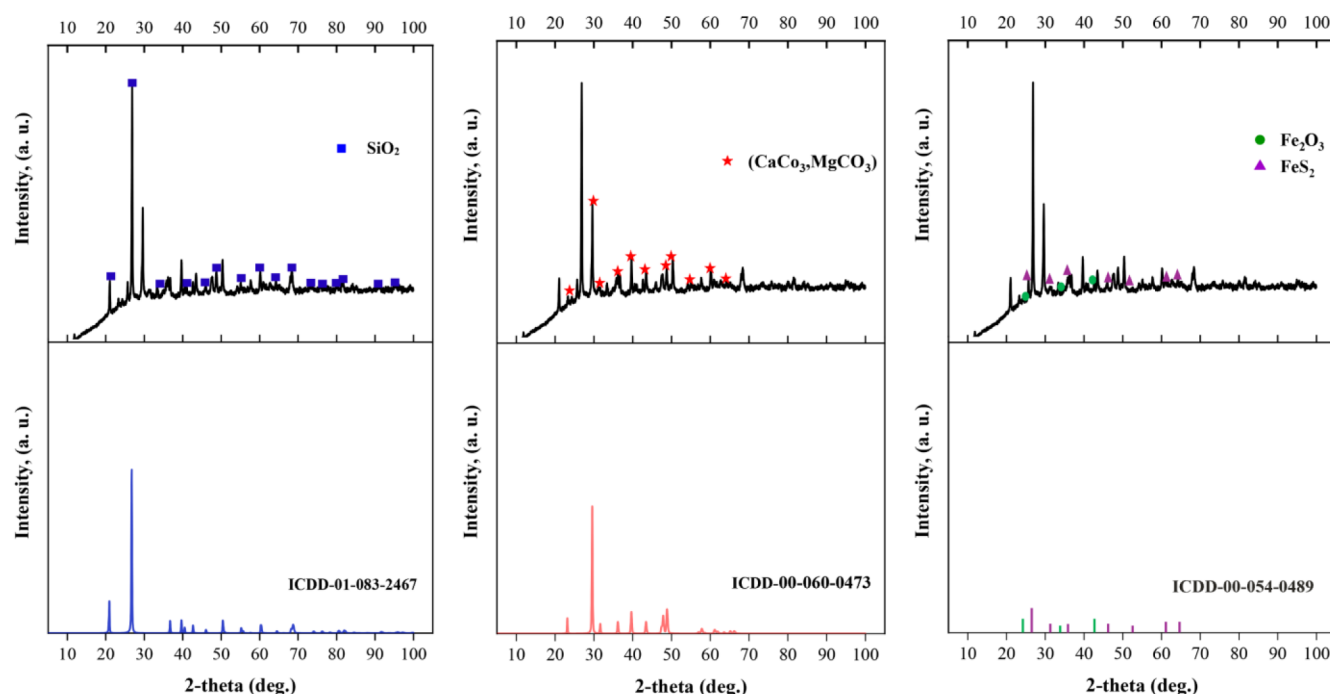


Figure 8—Powder XRD patterns of solid products after hydrothermal treatment in the presence of catalyst, match with ICDD cards number

Conclusion

This study investigates the efficiency and performance of iron-based catalyst (iron (III) nitrate) to improve the *in-situ* hydrothermal treatment of Domanik low-permeable shale rocks. Experimental results demonstrated substantial enhancements in the quantity and quality of synthetic oil during catalytic hydrothermal conversion. The key findings are as follows:

- The catalyst increased synthetic oil yields by 2.56% at 300°C, 5.39% at 350°C, and 4.87% at 400°C compared to experiments conducted without catalyst.
- The presence of catalyst led to a decrease in hydrocarbon gases (C_1 – C_4) and an increase in non-hydrocarbon gases, especially CO_2 and H_2S .
- Analytical results (GC, GC-MS, SARA) demonstrated a higher content of light n-alkanes (C_9 – C_{20}) and aliphatic hydrocarbons in the generated synthetic oil.
- The obtained synthetic oils were characterized by a higher H/C ratio and a notable reduction in sulfure content.
- XRD analysis indicated that the presence of iron sulfides (Fe_3S_2) and iron oxides (Fe_2O_3) plays a significant role in facilitating the effective desulfurization of bitumen and kerogen.

The findings provide valuable insights for optimizing *in-situ* conversion and advancing the sustainable exploitation of unconventional hydrocarbon resources.

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